



Bonsa Adugna

Software Engineer / ML Engineer

CONTACT

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📍Addis Ababa, Ethiopia

TECHNICAL SKILLS

LANGUAGES

Python C C++ Java JavaScript Dart Rust SQL

FRAMEWORKS

Django REST FastAPI Next.js React Flutter Flask Express.js

ML / AI

PyTorch scikit-learn Hugging Face RAG FAISS Pinecone
NumPy Pandas

DATABASES

MySQL MongoDB PostgreSQL SQLite

CLOUD / DEVOPS

AWS GCP Docker Git CI/CD Linux

DESIGN

Figma Adobe XD Canva Wireframing Prototyping

LANGUAGES

English Professional
Amharic Native

EDUCATION

B.Sc. Computer Science

Adama Science and Technology University

CGPA: 3.63 / 4.0

2022 — 2026

Real-Time Backend Systems

EPIC Institute of Technology

Distributed Systems, Raft, Fault-Tolerant Design

2025 — 2026

ACHIEVEMENTS

- ACPC (African & Arab Collegiate Programming Championship) Finalist
- Generative AI for Africa Hackathon Semi-Finalist
- Expert Rating on Codeforces — Top 5% globally
- 800+ algorithmic problems solved on LeetCode
- Best Lecturer at A2SV among 8 educators

PORTFOLIO

🌐github.com/adugna
🌐linkedin.com/in/adugna
🌐adugna.dev
🌐behance.net/adugna
🌐dribbble.com/adugna
🌐figma.com/@adugna

PROFESSIONAL SUMMARY

Software Engineer and Machine Learning Engineer with deep expertise in backend systems, distributed architectures, and AI/ML pipelines. Proven ability to design fault-tolerant systems, optimize large-scale data pipelines, and build production-grade ML applications. Combines strong algorithmic foundations with practical engineering, delivering scalable solutions from concept to deployment. Experienced in competitive programming, system design, and cross-functional team leadership.

PROFESSIONAL EXPERIENCE

Software Engineer

Robotic Games Inc. (Funded by Founders) — San Francisco, CA

2025 — Present

- Applied SimHash-based deduplication with adaptive Hamming thresholds, reducing redundant analyses by 60-80% using Python and Redis.
- Created a content-matching pipeline on AWS (FastAPI, PostgreSQL), lowering API costs by 60% and reducing median latency from 2.3s to 0.8s.
- Developed a browser extension (React, WebExtensions API) filtering unwanted web content in real-time, cutting scan time to under 20ms.
- Engineered a low-latency backend (<370ms) with FastAPI and Redis, integrating an on-device privacy layer for secure processing.

Head of Education

Africa to Silicon Valley (A2SV) — Adama, Ethiopia

2023 — 2024

- Mentored 60+ students in software development and data structures; organized 19 contests and authored solutions.
- Resulted in 22 internships and job placements for students in top tech companies.
- Recognized as Best Lecturer among eight educators for instructional quality and student outcomes.

Project Lead — Sparck (AI Social Media Platform)

Eskalate — Palo Alto, CA

2025 — Present

- Led a 15-member team to develop Sparck, a social media management platform with 98.9% uptime.
- Created 50+ RESTful APIs with OAuth2/JWT authentication, rate limiting, and cursor pagination.
- Reduced feature lead time by 30% via query optimization, Redis caching, and connection tuning.
- Containerized services with Docker and automated CI/CD pipelines using GitHub Actions.

Software Engineering Intern

Eskalate — Palo Alto, CA

2024

- Developed socket-based chat features supporting 500+ concurrent users with zero-downtime releases.
- Improved code quality by driving prototyping, testing, and reviews, increasing test coverage from 55% to 85%.

DEVELOPMENT PROJECTS

📦 Career Pilot — AI Career Guidance Platform

2024

AI-powered career guidance platform serving 280+ monthly active users with personalized recommendations.

Next.js Frammer Motion FastAPI OAuth 2.0 RBAC

- Designed responsive frontend with Next.js and Frammer Motion for seamless user experience.
- Established robust authentication with OAuth 2.0 and role-based access control, zero unauthorized access incidents.
- Integrated FastAPI microservices to serve personalized career recommendations at scale.

GitHubLive Demo

📦 QuizMe — Adaptive AI Quiz Platform

2024

AI-driven adaptive quiz platform processing 80+ quizzes daily with RAG pipelines and vector search.

Python Django LangChain FAISS Pinecone RAG

- Engineered RAG pipelines with vector search and prompt orchestration for adaptive question generation.
- Ranked Top 30 among 1,000+ teams in a national hackathon for technical innovation.
- Delivered full-stack prototype using Django, LangChain, and FAISS/Pinecone vector databases.

GitHubLive Demo

UI/UX DESIGN PROJECTS

🔗 Sparck AI Social Media Dashboard

Dashboard

Goal: Design an intuitive social media management dashboard for content creation and analytics.

Figma Adobe XD

- Content calendar with drag-and-drop scheduling and multi-channel publishing
- Analytics overview with engagement metrics and audience growth charts

Outcome: Streamlined workflow reducing content scheduling time by 40% for social media managers.

🔗 Career Pilot UI/UX

Web App

Goal: Create a clean, trustworthy interface for AI-powered career guidance.

Figma Canva

- Personalized career roadmap with milestone tracking and skill gap analysis
- Job matching interface with filterable listings and application tracker

Outcome: Intuitive design achieving 85% task completion rate in usability testing sessions.

🔗 QuizMe Mobile Interface

Mobile App

Goal: Design an engaging, adaptive quiz experience for mobile learners.

Figma Adobe XD

- Gamified quiz flow with progress bars, streak counters, and achievement badges
- Adaptive difficulty UI that responds to user performance in real-time

Outcome: Increased daily active users by 45% through gamification and smooth interactions.